Assessing the environmental impact of new Circular business models

Conference "New Business Models" - Exploring a changing view on organizing value creation – Toulouse, France, 16-17 June 2016

Nancy Bocken[1,2,*], Karen Miller[3] and Steve Evans[1]

[1] *University of Cambridge, Department of Engineering, Institute for Manufacturing, 17 Charles Babbage Road, Cambridge CB3 0FS, United Kingdom*
**nmpb2@cam.ac.uk*

[2] *Delft University of Technology, Industrial Design Engineering (building 32), Delft University of Technology, Landbergstraat 15, 2628 CE Delft, The Netherlands*

[3] *University of Brighton, Mithras House, Lewes Road, Brighton, BN2 4AT, United Kingdom.*

Abstract

Circular business models are increasingly viewed as a key driver for business competitiveness and sustainability. Yet it is unclear what the positive (or negative) environmental impact of such new business models might be. Existing methods such as Life Cycle Assessment and Mass Flow Analysis, typically more suited for product-level and material-level environmental assessment respectively, may be modified to assess the impact of new circular business models. However, these might be too time and resource-intensive to effectively guide decision-making. Building on earlier work on circular design and business models, the performance economy and life cycle and systems thinking, this work proposes a list of guiding principles to start assessing the impact of new circular business models. The guiding principles are organised according to the following high-level strategies: Slowing effects; Closing effects; Life cycle effects, and Systems effects. Slowing effects are linked to long lasting products and extending product life as part of the new business model. Closing effects relate to the potential for recycling, or product life post-consumer use, as a result of the circular business model innovation. Life cycle effects are the effects not yet captured within these slowing and closing effects, which arise across the product life cycle from raw material sourcing up until production, transport, use and disposal of products. Finally, Systems effects are the wider impacts of circular business model innovation. For each high-level strategy key questions were formulated to assess the environmental impact reduction of the new circular business model.

It was found that the strategies and guiding principles help stimulate debate on the environmental direction of new circular business models. This work has provided insights to guide a potential direction for assessing the environmental impact of new circular business models, and added to the sparse body of literature in this field.

Keywords: *Circularity assessment; circular business; sustainability assessment; sustainable business; circular design; streamlined LCA.*

1. Introduction

The Circular Economy is increasingly viewed as a key driver for business competitiveness and future sustainability under increasing resource scarcity. The switch from the current linear economy to a circular one has recently attracted attention from major global companies and policymakers (Lewandowski, 2016). The Ellen MacArthur Foundation (2015), whose mission is to accelerate the move to a circular economy, regards the circular economy as essential in reshaping resource utilization and a resilient economy.

Scholars view the circular economy as a key driver for sustainability; however, to enable this transition, business models, product design and supply chains need to change (Rashid et al., 2013; Bocken et al., 2016). Business models are viewed as a key lever to change to business as usual (Chesbrough, 2010) and can be an important driver for business sustainability (Bocken et al., 2013). However, the literature on the potential impact of such new business models, with the exception of conceptual work by Tukker (2004; 2015) for example, is yet sparse. Currently, it is unknown what the potential positive (or negative impact) of such new business models could be.

This paper addresses the following research question: *How can the environmental impact of circular business models be assessed?*

Building on earlier work on circular design and business models (Bocken et al., 2016), the performance economy (Stahel, 2010) and life cycle and systems thinking, this work proposes a list of guiding principles to start assessing the impact of such new circular business models. This paper first reviews the literature on circular business models and environmental assessment. Subsequently a brief illustrative case of the proposed framework of circular guiding principles is developed. Finally, suggestions for future research in the area of circular business models are developed.

2. Literature on environmental assessment and circular economy

This section contains a brief overview of literature on circular business models, relevant

environmental assessment methods and the research gap.

2.1 Circular business models

Circular business models first optimise the value of products through reuse and long product lifetimes, and second enable closed loop recycling with a viable business case. In a circular economy, products are designed for multiple uses or even lifecycles (Rashid et al., 2013). Stahel (1994, p. 179; 2010), referred to “closed loop systems” and distinguished two fundamentally different types of loops within a closed loop system: (1) reuse of goods, and (2) recycling of materials. Building on this, Bocken et al. (2016) define three strategies for the move towards a circular economy: Slowing loops (reuse), closing loops (recycling) and narrowing loops (efficiency) (see Figure 1).

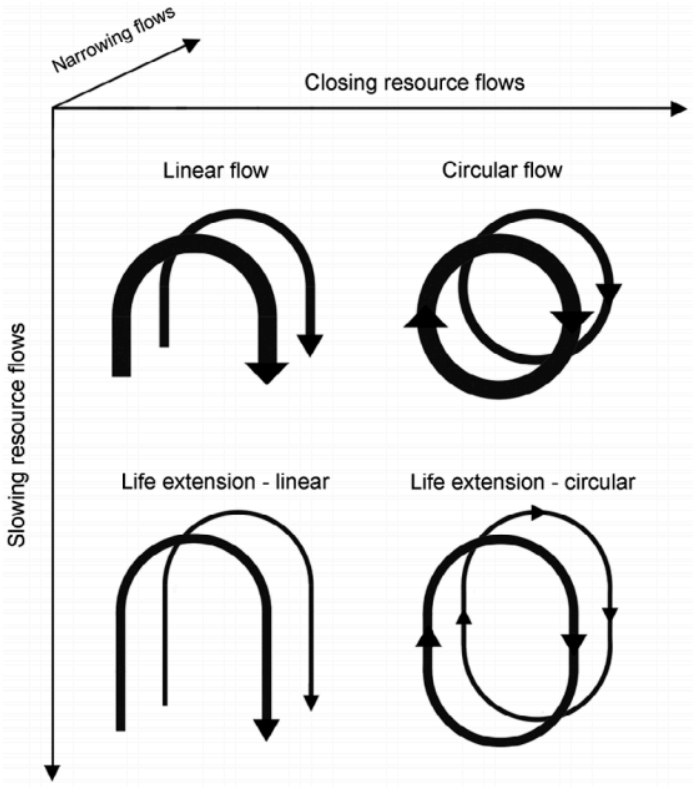


Figure 1. Framework of closing, slowing and narrowing resource loops. Bocken et al. (2016)

Whereas ‘narrowing loops’ (e.g. lightening products and manufacturing efficiencies)

already takes place in a linear economy, a lot of work still needs to be done to slow and close resource loops.

Circular business models may be viewed as a subset of 'sustainable business models' (Stubbs & Cocklin, 2008), which take into account multiple stakeholder concerns, in particular Society and Environment in the way that business is done and performance is defined. However, it is not currently clear what the potential impact of such new business models may be and how these might contribute to sustainability.

2.2 "Circular" assessment methods

Environmental assessment tools may be highly qualitative or quantitative as explained according to the classification in Figure 1. Existing quantitative methods such as Life Cycle Assessment and Mass Flow Analysis have been developed and used to assess product-level and material-level environmental assessment respectively. Life Cycle Assessment (LCA) is an environmental assessment method, which may assist in identifying opportunities to improve the environmental performance of products, inform decision makers, select environmental indicators of environmental performance, and support marketing statements (ISO 14044, 2006). Mass Flow Analysis (MFA) is a method of investigating flows of specific materials through an economic system in a specific geographic area during a certain period of time (Kytzia et al., 2004).

LCA typically incorporates the end of life stages (e.g. recyclability), whereas MFA allows for an overview of a 'system' of specific flows of materials over time. Indeed, whereas MFA is often used to address issues around the water-food-energy nexus, environmental and human health and economic systems (e.g. material flows in cities), LCA analysis is typically used to look at product chains as well as user behaviour (Blomsma, 2015). The Circularity Canvas by Blomsma (2015) brings together the notions of product (re) use and flows of materials in a Sankey diagram depicting resource flows.

However, in the early phases of business model design, similar to product design, specific design parameters are yet unknown. Hence, it is beneficial to integrate

environmental aspects in order to understand environmental impacts right from the start, when the greatest impact may potentially be achieved, and parameters altered to derive the greatest environmental benefit (Bocken et al., 2014). While this phase is largely uncertain and the details of design still need resolving, it is argued here that rules of thumb and guidelines (Figure 2) will be most beneficial.

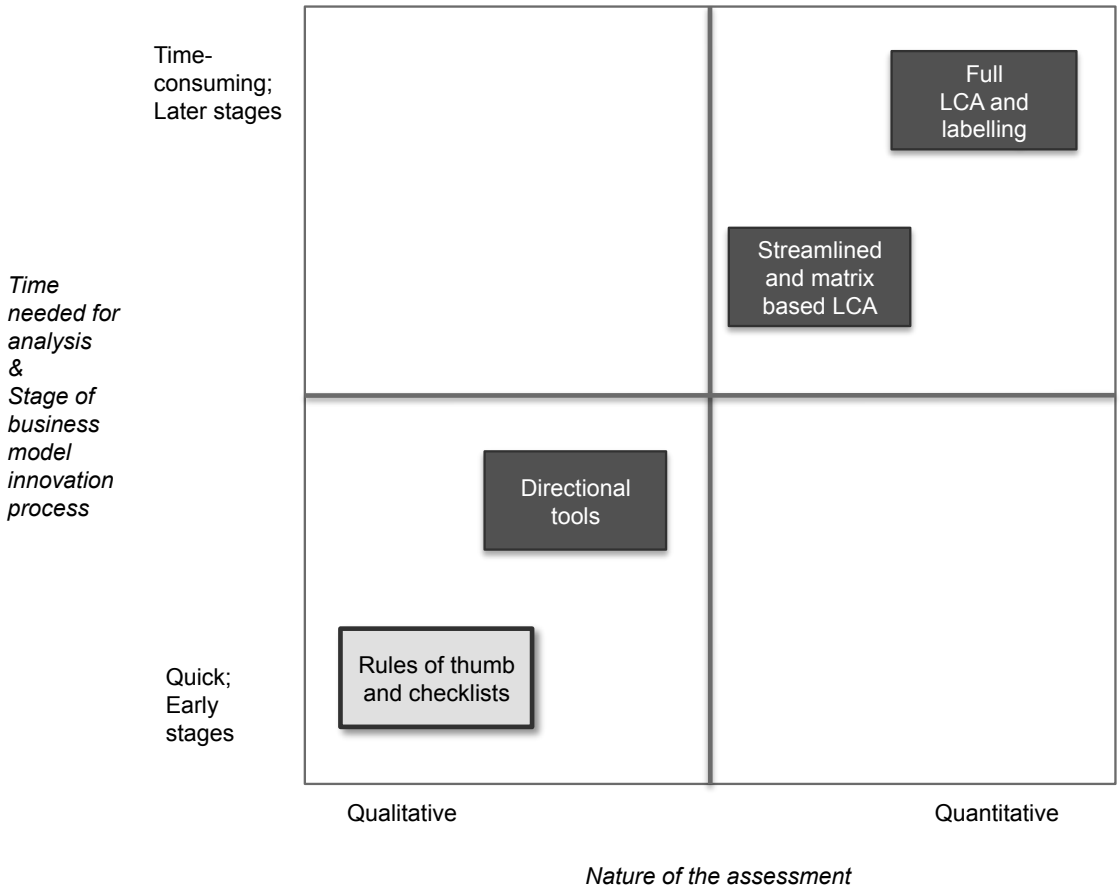


Figure 2. Classification of different types of product-level environmental impact assessment tools
Source: Developed from: Bocken & Allwood (2012) adapted from Forum for the Future (2010)

2.3 Research gap

While the work on a Circular Economy has received increasing interest (Rashid et al., 2013) there is still substantial uncertainty on the potential impact of new circular

business models. Few if any suitable tools have been established on circularity assessment; and even fewer for circular business models specifically. One exception is the “Circularity Canvas’ by Blomsma (2015), which can help visualise resource flows in a business. However, this canvas is focused more on mapping current resource flows than guiding future strategic business model design. This paper proposes a simple method to guide the development of new circular business models in order to achieve the greatest environmental benefit when design parameters are still flexible.

3. Method

The proposed method was developed based on simple and fundamental ‘circularity strategies’ identified from the literature (Stahel, 1994, p. 179; 2010; and Bocken et al., 2016).

Stahel (1994, p. 179; 2010) distinguished two fundamentally different types of loops that support the move to a circular economy: (1) reuse of goods, and (2) recycling of materials. Based on these, Bocken et al. (2016) defined three strategies for the move towards a circular economy: slowing loops (reuse), closing loops (recycling) and narrowing loops (efficiency).

As a result, the following high-level strategies were developed: Slowing effects; Closing effects; Life cycle effects, and Systems effects.

In line with Stahel, Slowing and Closing effects are most important in a circular economy (Stahel, 1994, p. 179; 2010). Slowing effects are linked to long lasting products and extending product life as part of the new business model (Bocken et al, 2016). Closing effects are in essence about assessing the potential for recycling, or what happens to products post-consumer use, as a result of the circular business model innovation (Bocken et al, 2016). Life cycle effects are the effects not yet captured within these slowing and closing effects, which arise across the product life cycle from raw material sourcing, up until production, transport, use and disposal of products. The idea of ‘life cycle effects’ is derived from LCA, which takes a product life cycle perspective (ISO

14044, 2006). Finally, Systems effects are the wider impacts of the circular business model innovation. MFA also takes a 'systems view', e.g. the flows of a material in an economic system (Kytzia et al., 2004). However, in this case, the effects might be the wider changes or impacts that the new circular business model might have on society or other businesses.

For each of these high-level strategies key questions were formulated to assess the environmental impact reduction or improvement of new circular business models. A set of circular guiding principles (i.e. key questions) was developed for each of these environmental strategies – slowing, closing, life cycle and systems strategies and effects. Table 1 includes the high-level circularity strategies.

Table 1. Circularity strategies developed in this research.

Slowing effects *(long lasting products and extending product life)*
Closing effects *(recycling)*
Life cycle effects *(effects across raw material sourcing, production, transport, use and disposal- not yet captured)*
Systems effects *(wider impacts of the innovation)*

This approach was further developed and tested through a case study in the clothing retail industry (specifically, a fashion retailer). The case company aims to reduce waste through new circular business models. While 'waste reduction' was the main metric, the improvement in waste reduction was not at the expense of other environmental metrics such as water usage and carbon emissions. The company wants to investigate the reduction in waste of new business models before allocating significant resources to implementing a certain business model.

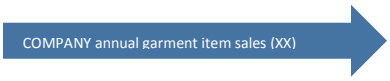
The aim of the approach developed in this research was to help companies in the move to a circular economy. The quick assessment method developed can help test the environmental viability of new circular business models.

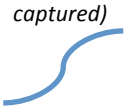
The approach was used in two instances by the case company: firstly in earlier phases of business model development and secondly as background material for a more in-depth business model development workshop.

4. Circular business models – rapid environmental assessment

Table 2 includes the Rapid Circularity Assessment (RCA) to assess the environmental impact of new circular business model ideas for clothing retailers.

Table 2. Rapid Circularity Assessment (RCA)

	Flows – XX items sold annually by COMPANY 	**Stocks — UK wardrobe of 6bn items or 100 items per person** 
Closing effects *(Recycling)* 	**Key questions to consider:** 1. Does it increase the *recyclability* of new COMPANY clothes being manufactured? (E.g. consider design for disassembly and keeping technical and biological materials separate) 2. Does it increase *recycling rates for* new COMPANY clothes being manufactured? (E.g. by ensuring that used product collection at the end of the useful product life time, a product deposit system, or leasing, where the material remains in the ownership of the company) Estimated impact: xxx reduction in waste Confidence: low, medium, high	**Key questions to consider:** 1. Does it increase the recyclability of the existing clothing stock? 2. Does it increase the number of clothing items from the UK wardrobe given away for recycling? 3. Does it have ability to reverse trends of textiles going to landfill? (E.g. by repurposing the materials and increasing recycling rates) Estimated impact: xxx reduction in waste Confidence: low, medium, high
Slowing effects *(Long lasting products and extending product life)* 	**Key questions to consider:** 1. Does it increase the useful lifetime of *new* garments (e.g. through maintenance and care, repair, re-manufacture, reuse)? 2. Does it reduce total *new* garment production? Estimated impact: xxx reduction in waste Confidence: low, medium, high	**Key questions to consider:** 1. Does it reduce the total number of items in the UK wardrobe? 2. Does it lead consumers to better use their *own* existing wardrobe? Will it make them buy less as a result? 3. Does it lead to the stock of clothing being used better *in general* (e.g. by others, UK, abroad)? 4. Does it have the ability to reverse trends of clothing "overconsumption"? Estimated impact: xxx reduction in waste Confidence: low, medium, high
Life cycle effects *(Effects across raw*	**Key questions to consider:** 1. Does it lead to any 'design efficiencies' (e.g. less material per product)? 2. Does it lead to any manufacturing efficiencies?	**Key questions to consider:** 1. Does it lead to any transportation savings for handling the current clothing stock? 2. Does it lead to more efficient or less cleaning for the

material sourcing, production, transport, use and disposal- not yet captured) 	(Consider material extraction/ production and garment production) 3. Does it lead to any transportation savings? 4. Does it lead to more efficient or less cleaning? 5. Does it lead to cleaner forms of recycling not captured yet? (E.g. less toxic, biodegradable)? 6. Does it lead to any other efficiencies not yet captured (e.g. sale stock/over ordering samples/fabrics, no ironing needed etc.) Estimated impact: xxx reduction in waste and other environmental categories Confidence: low, medium, high	current clothing stock? 3. Does it lead to any other efficiency/ savings (e.g. ironing, tumble drying) not yet captured for current clothing stocks? Estimated impact: xxx reduction in waste and other environmental categories Confidence: low, medium, high
Systems effects *(Wider impacts of the innovation)* 	**Key questions to consider:** 1. Does it lead to negative rebound effects (e.g. consuming more) or positive ones (e.g. from fast to slow fashion)? 2. Are there any unintended consequences by upcycling the value of waste (e.g., where textiles are used as a low cost filler what is likely replacement if we can divert textile waste)? 3. Does it lead to radical changes for stores and employees? What are the impacts on those stakeholders and others? Who are the potential winners/ losers? 4. What is the effect of multiple co-existing business models? 5. What is the impact on society of this innovation? 6. What is required for other companies to follow – is that considered in this new business model? Estimated impact: xxx reduction in waste and other sustainability categories - other positive and negative impacts Confidence: low, medium, high	

Note. These are guiding questions to facilitate your analysis. Please add as required to analyse your ideas. The UK garment stocks data of 6bn items are retrieved from WRAP (2012).

The assessment is organised according to the effects in Table 1 – Slowing, closing, life cycle and systems effects. In addition, it was recognized that some innovations could have effects on the flows of new materials (left column) of products sold by the company, while others could help improve the usage of the existing clothing stock (right column; clothing already in circulation, but may remain unused). WRAP UK for example estimates that the average UK household owns around £4,000 worth of clothes – and around 30% of clothing in wardrobes has not been worn for at least a year (WRAP, 2012). The assessment method allows companies to go through each of the questions step-by-step to assess their business models.

This method has been used by the case company in two ways:

1. The strategies in Table 1 have been used in the early stages of business model development when changes could still be made. It helped the company identify business models that slow and close loops.

2. The strategies in Table 2 have been used as background material for an in-depth business model design workshop. They were used to support further business model development and refinement.

Through usage in these two ways, it was found that the environmental strategies and guiding principles could help stimulate debate on the environmental direction of new circular business models. They can help improve the environmental performance of such new business models at the early stages of business model innovation.

5. Conclusion and discussion

The Rapid Circularity Assessment (RCA) was developed to support companies in circular business model innovation to help them consider the most environmentally beneficial options and make changes to the design while these are feasible. Through the case study there are early indications that this method is beneficial. However, significant further testing and development is required. Nevertheless, the guiding principles could serve as an input to more detailed LCA or MFA work. For example, it can help build the assessment framework and assumptions for a detailed LCA (ISO 14044, 2006). It can also help focus the analysis and draw boundaries for MFA.

This work has provided insights in a potential direction for assessing the environmental impact of new circular business models and contributed to the scant body of literature in the field. The RCA method provides an indication on whether the business model is going in a fruitful direction before committing excessive resources and embarking on deeper modelling. Future work could focus on environmental assessment methods across the full spectrum illustrated in Figure 2 – from rules of thumb up until highly detailed quantitative assessment. Finally, more circular business models are needed in practice to be able to assess their impact after implementation and guide future business model development.

Acknowledgements

Project funding was secured through InnovateUK's funding call "Supply chain innovation towards a circular economy" (InnovateUK, 2013).

Short bio of the authors

Dr Nancy Bocken is an associate professor at TU Delft, Industrial Design Engineering. She was awarded the TU Delft Technology Fellowship to pursue her research in sustainable business models, and innovation for a circular economy. She has an appointment at University of Cambridge as a senior research associate and is Fellow at the Cambridge Institute for Sustainability Leadership. While her broader interest is closing the "idea-action gap" in sustainability, her research focus is on sustainable business model innovation. Nancy holds a PhD from the University of Cambridge, funded by Unilever. In the past, she worked for DHL, Accenture, and ING.

Dr Karen Miller is a Senior Lecturer in Business Studies for Fashion & Textiles and Fashion Communications at the University of Brighton. Additionally, she is a Visiting Industrial Fellow for Industrial Systems Manufacturing & Management at the Institute for Manufacturing, University of Cambridge. At Cambridge she is also conducting research with the Centre for Industrial Sustainability on developing more sustainable approaches in fashion retailing. Her main research interests are leadership for design and innovation, and sustainable business development in fashion.

Prof Steve Evans spent 12 years in industry, finally as Engineering Systems Manager at Martin-Baker Engineering, the world-leading manufacturer of ejection seats. He became Professor of Life Cycle Engineering at Cranfield University in 1998 and joined the University of Cambridge Institute for Manufacturing in 2011. He is Director of the national EPSRC Centre in Industrial Sustainability, with over 50 staff. His research seeks a deep understanding of how industry brings environmental and social sustainability concerns into its design and manufacturing practices, with an emphasis on urgent and practical change now and system level change that offers hope for a sustainable future.

References

Blomsma, F. 2015. Navigating the circular economy: the particle-versus-product debate. Global Cleaner Production & Sustainable Consumption Conference, 1 - 4 November 2015, Sitges, Barcelona, Spain.

Bocken, N.M.P., de Pauw, I., van der Grinten, B., Bakker, C. 2016. Product design and business model strategies for a circular economy. Journal of Industrial and Production Engineering, 32 (1), 67-81.

Bocken, N., Allwood, J. 2012. Strategies to reduce the carbon footprint of consumer goods by influencing stakeholders. Journal of Cleaner Production, 35, 118-129.

Bocken, N., Farracho, M., Bosworth, R., Kemp, R. 2014. The front-end of eco-innovation for eco-innovative small and medium sized companies. Journal of Engineering and Technology Management, 31, 43–57.

Bocken, N., Short, S., Rana, P., Evans, S. 2013. A value mapping tool for sustainable business modelling. Corporate Governance, 13 (5), 482 – 497.

Chesbrough, H. 2010. Business Model Innovation: Opportunities and Barriers. Long Range Planning 43, 354-363.

Ellen McArthur Foundation. 2015. Home page. Available at: https://www.ellenmacarthurfoundation.org/ (accessed 27 May 2016).

Forum for the Future. 2010. Metrics. Retrieved from the WWW December 2010: http://www.forumforthefuture.org/projects/metrics

INNOVATEUK. 2013. *Supply chain innovation towards a circular economy. Available at:* https://interact.innovateuk.org/competition-display-page/-/asset_publisher/RqEt2AKmEBhi/content/supply-chain-innovation-towards-a-circular-economy#: UK Government (Accessed 27 May 2016).

ISO 14044. 2006. Environmental management — Life cycle assessment — Principles and framework. International Organization for Standardization. 2nd ed.

Kytzia, S., FAist, M., Baccini, P. 2004. Economically extended—MFA: a material flow approach for a better understanding of food production chain. Journal of Cleaner Production 12, 877–889.

Lewandowski, M. 2016. Designing the Business Models for Circular Economy—Towards the Conceptual Framework. Sustainability, 8(1), 43; doi:10.3390/su8010043.

Rashid, A., Asif, F., Krajnik, P., Nicolescu, C. 2013. Resource Conservative Manufacturing: an essential change in business and technology paradigm for sustainable manufacturing. Journal of Cleaner Production, 57, 166-177.

Stahel, W. R., 1994. "The utilization focused service economy: Resource efficiency," in B. R. Allenby and D. J. Richards (eds), The Greening of Industrial Ecosystems, National Academy Press, Washington, DC, 178–190 (1994).

Stahel, W. R. 2010. The performance economy, Palgrave Macmillan Hampshire, UK.

Stubbs, W., Cocklin, C. 2008. Conceptualizing a "sustainability business model". Organization & Environment, 21(2), 103-127.

Tukker, A., 2004. Eight types of product-service system: eight ways to sustain- ability? Business Strategy & the Environment, 13 (4), 246-260.

Tukker, A., 2015. Product services for a resource-efficient and circular economy– a review, Journal of Cleaner Production, 97, 76-91.

WRAP. 2012. WRAP. Valuing our clothes. Available at: http://www.wrap.org.uk/content/valuing-our-clothes (Accessed 27 May 2016)